



GE Medical Systems

Technical Publication

**Direction 2322593-100
Revision 0**

GE Medical Systems Replacement of Gantry Reset Switch Bulb

**Service Note
T-1443
(Supercedes T-1415)**

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Revision History

REV	DATE	REASON FOR CHANGE
Rev 0	10/25/01	Initial Release

List of effective pages

Page	Rev.	Page	Rev.	Page	Rev.
All	0				

Replacement of Gantry Drives Reset Switch Bulb

Purpose

Increase reliability and reduce service cost regarding incandescent lamps part number 2138614 or 2158609. These lamps are used inside the Gantry Power Reset Switch.

Effectivity

All CT Z-series, RP, CT/i and QX/i systems having Gantry mounted Controls.

Problem

Bulb, part number 2138614, contained inside of the Gantry Power Reset Switch exhibits an excessive failure rate. Cause of these failures has been determined to be shock resulting from switch actuation. The reset switch is a spring-loaded push button that returns to its "at rest" position via spring pressure after being actuated.

Service Note T-1415 provided instructions for replacement of bulb 2138614 with a more reliable, shock resistant, bulb: part number 2158609.

This Service Note supercedes T-1415 and provides for replacement of both bulbs with a new LED: 2282712.

The old bulbs can be replaced with the new 2282712 LEDs upon failure or anytime during service. When ordering new parts, be certain to specify quantity 2 of part number 2282712 to receive the new LEDs.

Note: If you are upgrading from the original bulbs (2138614) or those specified in Service Note T-1415 (2158609) you must replace both bulbs with the LEDs.

Procedure

1. First, determine which style PDU is used in the system:
 - A. CT Compact PDU, model 2133533 - Proceed to step 2 below.
 - B. GLOBAL PDU, model 2113764 (-34) - Proceed to step 2 below.
 - C. HSA PDU, models 46-296220G1, 327106G1 - Proceed to step 7 below.

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2. Perform a system "Shutdown" at the CT console.
3. When the system has completed shutdown, turn-off the mains supply to the PDU using proper lock-out / tag-out procedures.
4. Modify the gantry lamp flasher circuit in the PDU as follows:
 - 1) **CT COMPACT PDU** - Open the top cover to gain access to the thumbscrews that secure the front cover and remove the front cover.
 - 1) Locate the PDU Control Board, 2139289. Note the version letter marked near the upper right corner of the board. If it is version "G" or greater no modification is necessary and you may proceed to step 0. If the board is marked version "A thru F" perform the following modification:
 - 2) Refer to figure 1. Locate resistors R183, R184, R185, and R186 near the lower left side of the board. Cut both leads of all four resistors to remove and discard them.

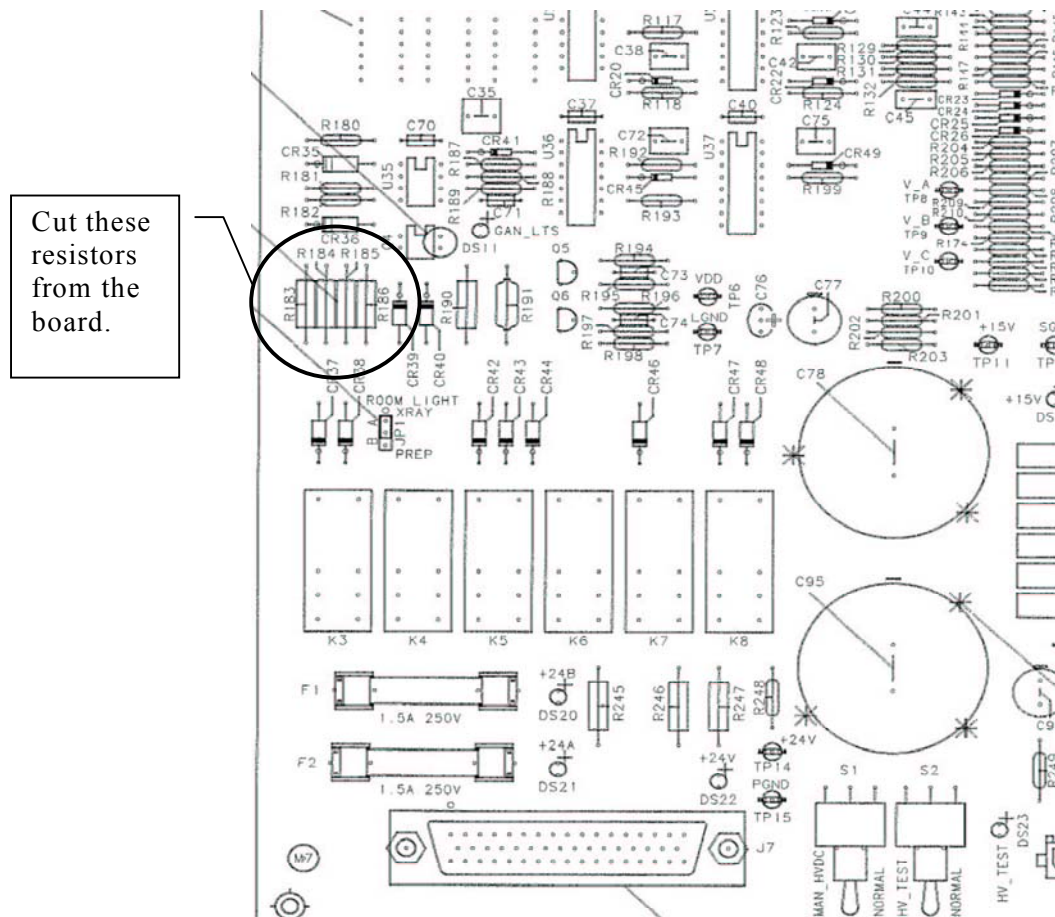


Figure 1 – Compact PDU Control Board Modification

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- 2) **GLOBAL PDU** - Open the front cover and remove the circuit breaker panel assembly cover on the lower left of the cabinet.
 1. Locate the Relay Control Board, 2113764-2 (On-Line Power #1625-142-01).
 2. Refer to figure 2. Locate R13, cut both leads to remove and discard it.

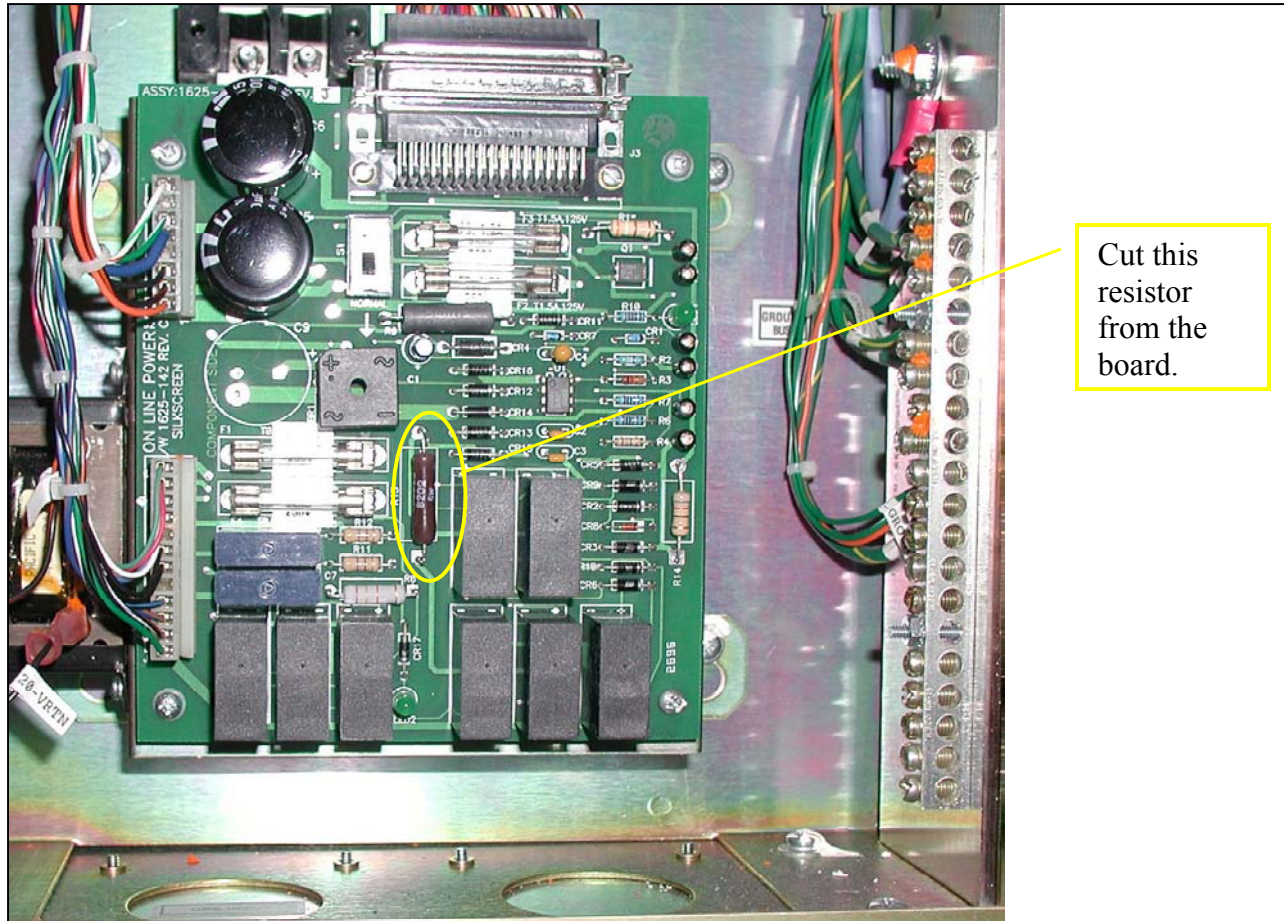


Figure 2 – Global PDU Relay Control Board Modification.

5. Reinstall all PDU covers.
6. Re-energize the system and allow applications software to load normally.
7. Reset gantry drives by depressing one of the gantry drives enable pushbuttons.
8. Refer to figure 3, below. Remove reset switch cap (part number 2114534 (right side) or 2114534-2 (left side)).

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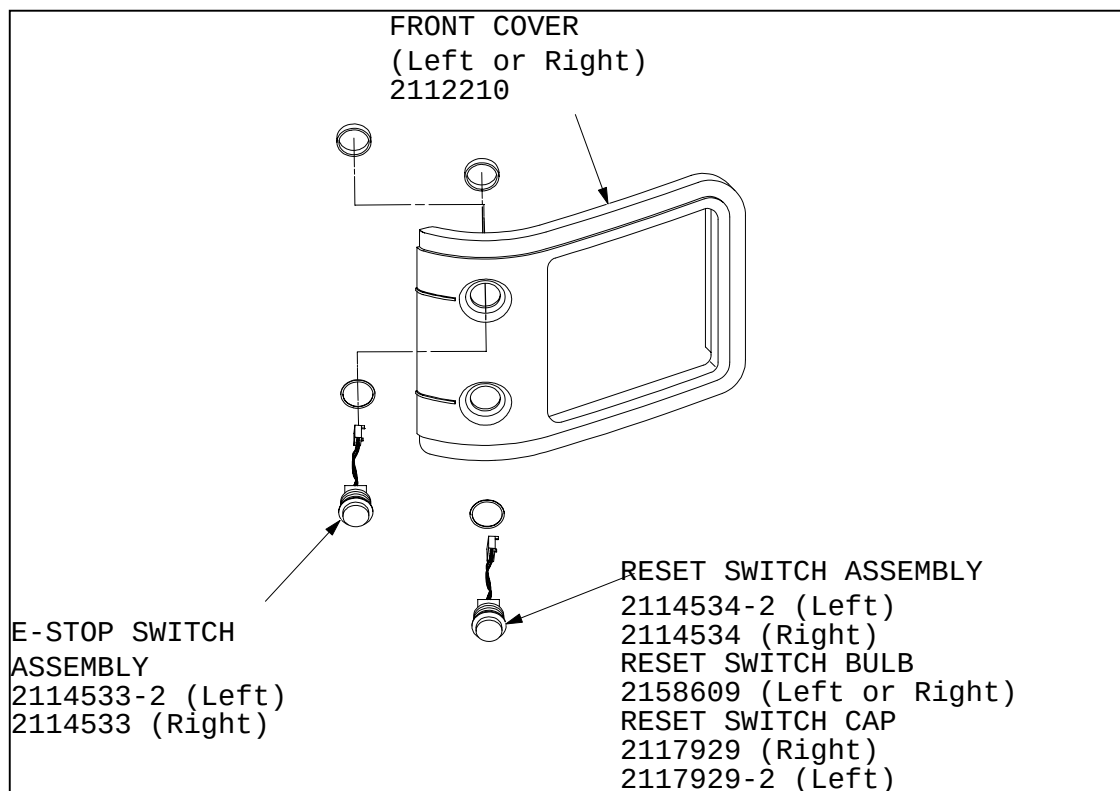


Figure 3 – Gantry Control Panel Assembly

9. Remove the existing bulb from the switch.
 10. Insert the LED into the switch socket. Using your forefinger, press the LED into place. The LED fits loosely in the socket and insertion may require more than 1 attempt to obtain positive seating.
- Note: If the LED does not light when inserted, you likely have it inserted with the wrong polarity. Remove the LED, turn it 180 degrees and reinsert it into the socket.**
11. Refer to figure 4. When the bulb has been seated, install the reset switch cap.

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Figure 4 - Completed Job.



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